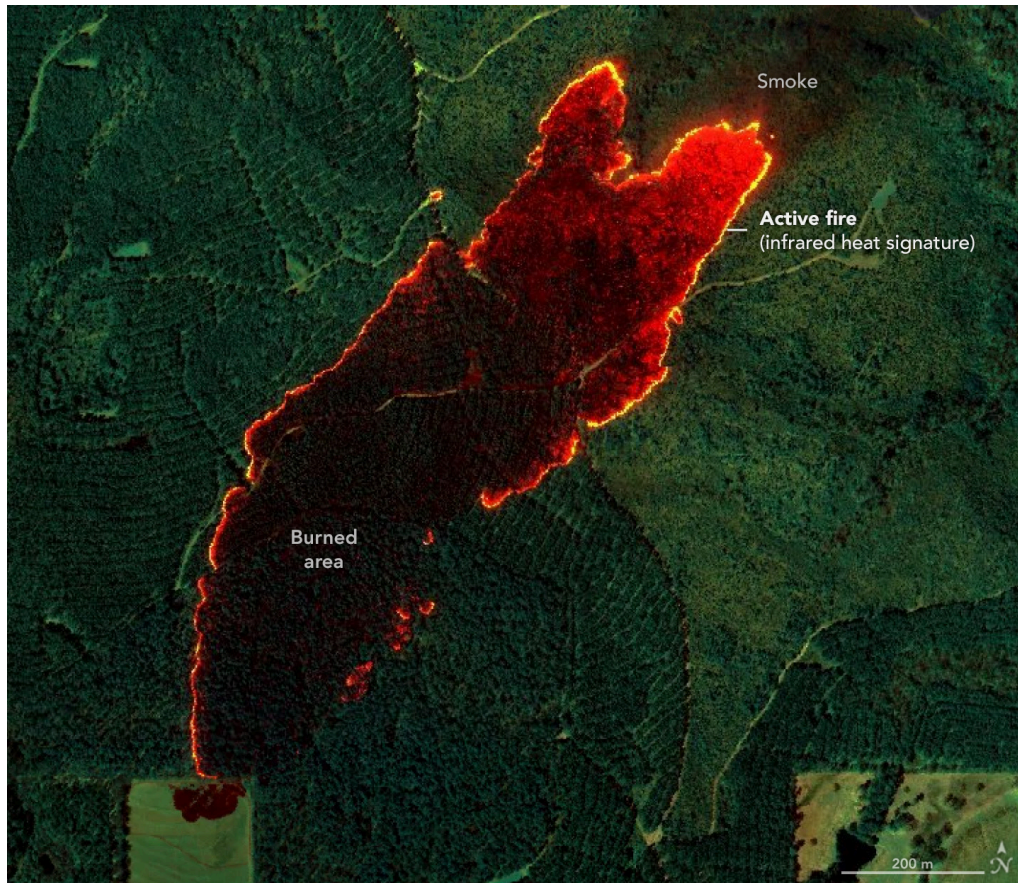


41068 ROBOTICS STUDIO 1

Bushfire Environmental Evacuation & Rescue (B.E.E.R.)

Component 1: Proposal Document



TEAM DETAILS

Group number: 25

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1. Team and Collaboration

1.1 Team Capabilities and Learning Goals

Team member	Relevant experience or existing skills	Skills they want to develop	Intended areas of contribution
Phu	Proficient in Python, C++, and Java, with hands-on experience using SolidWorks for CAD design and ROS2 for robotics development. Strong teamwork and communication skills developed through past collaborative technical projects, enabling effective coordination across both software and hardware aspects of a project.	Develop skills in the ROS2 CORE and virtual simulation, which extends to virtual physics, virtual world design, as well as robotic design and implementation.	Mainly the physics behind robotic movement within the virtual environment but also wherever else I am required
Taj	I have previous experience with communication and team management working in team projects at uts in previous semesters, as a leader and as a regular member. From previous classes I have used python and C++ as well as some experience with ROS2.	Further development of general coding skill as well as a deeper understanding of ROS2.	Wherever needed by the team and the project.
Faiyad	Experience with multiple programming languages, including Python, C++, C# and Java. Familiar with ROS 2 and RViz, as well as CAD design using SolidWorks.	I hope to further develop my skills in team communication and collaboration, ROS 2, RViz, system integration, and practical robotics development.	I am happy to contribute across all areas of the project, particularly software development, ROS 2, CAD/mechanical design, system integration, testing, troubleshooting and documentation.
Jack	Semi proficient in python and C++ through experience with other coding and robotics projects. Also strengths in team collaboration and communication to achieve a united goal.	I lack experience with ROS2 which I believe is essential in my engineering career. This will be my main focus area this semester.	Wherever I'm needed.

1.2 Team Operating Agreement

Team Expectations

- Attend weekly meetings
- Complete assigned tasks before internal deadlines
- Respond to team communication within 48 hours
- Contribute meaningfully to all group deliverables

Work Contribution & Accountability

- Tasks tracked via shared Kanban board
- Missed deadlines must be communicated in advance
- Repeated lack of contribution will trigger the conflict process

*If contribution remains insufficient after internal discussion, the issue will be escalated to the coordinator with documented evidence.

Conflict Resolution Process

1. Internal discussion
2. Documentation of issue
3. Escalation to subject coordinator (if unresolved)

Academic integrity

- All work must be original
- No plagiarism
- Each member must be able to explain their subsystem

Commitment Statement

We agree to work collaboratively, professionally, and ethically to deliver a high-quality system.

2. Project Direction

2.1 Candidate Applications

Suggested length: 50 words per use case

Identify 5 possible applications that align with the Project Vision outlined in the *Project Brief* document. These may be practical or speculative, but they should respond to a meaningful forest management or environmental monitoring need. Get creative!

Candidate application	Intended user or stakeholder	Need or potential value	Main technical challenge	Feasibility this semester
Habitat Analysis	National park rangers, researchers	Automate search for endangered or invasive plant species for the goal of faster and more targeted bush regeneration and protection.	Ability to identify a wide range of plant species reliably and efficiently	MEDIUM
Bushfire Environmental Evacuation & Rescue (BEER)	Firefighters, SES	Deployable automated robot to help identify escape routes and lead firefighters when they become trapped in a bushfire. Considering fire intensity, size of firetruck and terrain difficulty the robot will find an optimal path to safety.	Assessing a wide range of variables simultaneously to identify the safest route efficiently.	HIGH
Bushfire priority identification	Firefighters, SES	Uses a variety of sensors like heat mapping sensors (IR), lidar, and carbon soot sensor that are attached to drones/huskies to determine which fires/areas are the hottest/highest chance of spreading.	Sensors for fire detection and accurate priority determination	MEDIUM
Seed Spreader	Farmers/Agriculture, forester, silviculturists	Increase population of diminishing/rare flora. Helps repopulate and increase the numbers of said flora.	Healthy soil detection and planting processes.	MEDIUM
Missing Human Rescue	SES, police, Private Investigator	Allowing the use of less resources and law enforcement officers for "Man hunt".	Ability to find well hidden and deceased bodies.	LOW

2.2 Chosen Application and Stakeholder Need

Bushfire Environmental Evacuation & Rescue (BEER)

In the event that a firetruck team becomes stuck/trapped/lost in a bushfire due to being encircled by fire, lost due to heavy smoke or similar, the firetruck team could deploy an automated robot to help find a safe path to a safe location. An efficient robot would be able to reduce risk of harm to the fire team from prolonged exposure to hazardous bushfire conditions by finding the quickest, least dangerous path. The robot would be equipped with sensing equipment not usually available to firetruck crews, making it appropriate for making its way through the smoke and fire. The robot would be aware that the path it finds must be suitable for the firetruck as well as itself, considering variables such as amount of space, heat and surface incline and checking these variables are within an acceptable range. This technology could be added to an existing firefighting fleet with minimal modifications allowing quick adoption by firefighting services. Additionally, this technology could be utilised by firefighting services for more than just rescue, the robot could be used for scouting out fires in buildings as well as bushfires and even provide fire suppression from within the fire without needing to endanger firefighters lives. This idea was selected over others as it was deemed to be the most interesting idea and because it has the most potential for additional advanced functions beyond the idea's basic functionalities. This project is feasible as the basic idea should be manageable based on our skills, with the potential for more advanced function additions to unlock the higher marks.

2.3 Related Work

Existing research, product or system	What it does	Relevant strengths or limitations	How it informs your project
SES Remotely Piloted Aircrafts https://statements.qld.gov.au/statements/100929	Search and rescue, disaster assessment, direct communication, emergency resupplies.	High efficiency compared to walking or driving. Health and safety concerns reduced. Not automated (human operated)	Technology exists in this field already. Automation is not yet used.
Ground Based Firefighting Robots https://www.hcrot.com/w/hat-robots-help-firefighters	These robots can safely enter burning buildings equipped with firefighting tools such as water jets and foam dispensers to put out fires which can be targeted with thermal imaging cameras. They are controlled remotely by an operator.	Heat resistance - can still function properly in high temperature environments Not Automated	A robot operating automatically in a high temperature environment such as a bushfire is entirely possible.
Firefighting Co-bots https://news.griffith.edu.au/2026/02/16/ai-powered-robot-vehicles-team-up-to-fight-fires/	Fighting fires with robot-human collaborations is currently being explored and researched. The focus is on robots powered by AI that are supervised by a human operator with the idea being that the AI being able to	Human - Robot collaboration is being looked into as a serious addition to firefighting practices. Still in the research stage- not fully	Using robotics for firefighting and firefighting related purposes is being seriously considered by those in the related fields which

	handle some tasks autonomously reduces the cognitive load and stress of the human operator.	implemented in the real world.	gives validity to our ideas.
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2.4 Long-Term Concept of Operations

In approximately five years, the system is envisioned as a coordinated autonomous bushfire monitoring platform used by organisations such as the NSW SES, fire services, and national park authorities. The system would operate in forests, mountains and bushland during high-risk fire conditions or active bushfire events.

A Husky Robot equipped with LiDAR, cameras, temperature sensors, and localisation systems would then be deployed to investigate priority locations in greater detail. Eventually finding the optimal path to rescue trapped civilians with human survival being the number one priority.

The system would autonomously analyse sensor data, determine the location and severity of detected hotspots, assess surrounding hazards, and prioritise areas requiring urgent attention. The robot will also select safe navigation paths and continuously update their route as environmental conditions change.

A human operator would supervise the mission through a central interface displaying robot locations, detected hotspots, risk levels, navigation paths, and live sensor information with an interactive GUI. The operator could assign new objectives or override autonomous decisions when required.

The system would provide emergency responders with faster and safer situational awareness, reduce the need for personnel to enter hazardous areas, and support more informed bushfire response decisions.

2.5 Concept of Operations – For 41068 Robotics Studio 1

At the start of the scenario, a SES truck is positioned at a simulated bushland location representing the last known position of a firefighting crew that has become disoriented or trapped due to heavy smoke and encroaching fire. The environment contains a mix of clear ground, vegetation, and simulated fire hazards, with a defined safe location (e.g. a fire trail or safehouse) set as the mission goal.

The SES truck has an onboard robot husky that will go ahead and lead the truck to the safety point. The husky uses onboard simulated sensors (LiDAR, camera, carbon soot, IR heat) to perceive its immediate surroundings, including static obstacles such as trees, dense bushland, and fire zones. As it moves through the environment, it continuously scans for obstacles that intersect its planned path and identifies simulated fire regions that must be avoided due to their associated risk.

Using this perception data, the system evaluates possible paths toward the goal location and selects a route that avoids detected hazards while the robot applies a predefined clearance margin around obstacles to approximate the space required by the following firefighting vehicle. If an obstacle or fire region blocks the current path, the robot re-plans a local detour and continues toward the goal.

As the robot moves, its position, planned path, and any detected hazards are displayed on a simple operator user interface (UI), allowing a human supervisor to monitor progress without needing to directly control the robot. The operator's role in this simplified scenario is limited to initiating the mission and observing progress, representing the firefighting crew that would follow the robot in the full-scale application.

The scenario ends when the robot successfully reaches the designated safe location, having autonomously navigated around all static obstacles and hazards in its path. Success is demonstrated by the robot completing the traversal without collisions, the user interface accurately reflecting the robot's real-time position and path, and the system correctly identifying and avoiding all simulated fire hazards placed in the environment.

2.6 Scope Boundaries and Assumptions

Category	Current definition
The system will...	Autonomously move toward a designated safe location, such as a safe house or a fire trail. When an obstacle or fire hazard block its path, the husky will locally avoid the hazard and continue toward the designated safe location where a traversable route exists.
The system will not...	Extinguish fires, carry loads, or physically transport/rescue people .
Environmental assumptions	The terrain is sufficiently traversable for Husky movement, and at least one safe path to the designated safe location exists.
Robot or sensor assumptions	The Husky's simulated sensors can reliably detect relevant obstacles and fire hazards within their defined sensing range, and localisation data is sufficiently accurate for navigation.
User or operator assumptions	The operator can initiate the mission and specify the designated safe location. The firefighting crew or vehicle is assumed to be capable of following the route identified by the Husky.
Simulation simplifications	Fire intensity, vegetation/tree density, terrain complexity and sensor behaviour will be simplified compared with real bushfire conditions. Dynamic fire spread may be implemented as a stretch feature.
Other important constraints	The system will operate entirely in simulation during 41068. MVP navigation will use local/reactive hazard avoidance rather than full dynamic global route optimisation. Performance is limited by the capabilities of the provided Husky platform and available simulated sensors.

2.7 Project Title

Bushfire Environmental Evacuation & Rescue (B.E.E.R.)

Our project title was chosen to clearly reflect the core purpose of the system, that being guiding trapped or disoriented SES crews to safety during bushfire emergencies. "Evacuation & Rescue" directly communicates the mission-critical function of the robot, while "Bushfire Environmental" grounds the project in its target domain and operating conditions. The acronym B.E.E.R. was selected as it is short, memorable, and easy for both technical and non-technical stakeholders to recall, while still mapping meaningfully onto the project's full name rather than feeling forced or unrelated to the task.

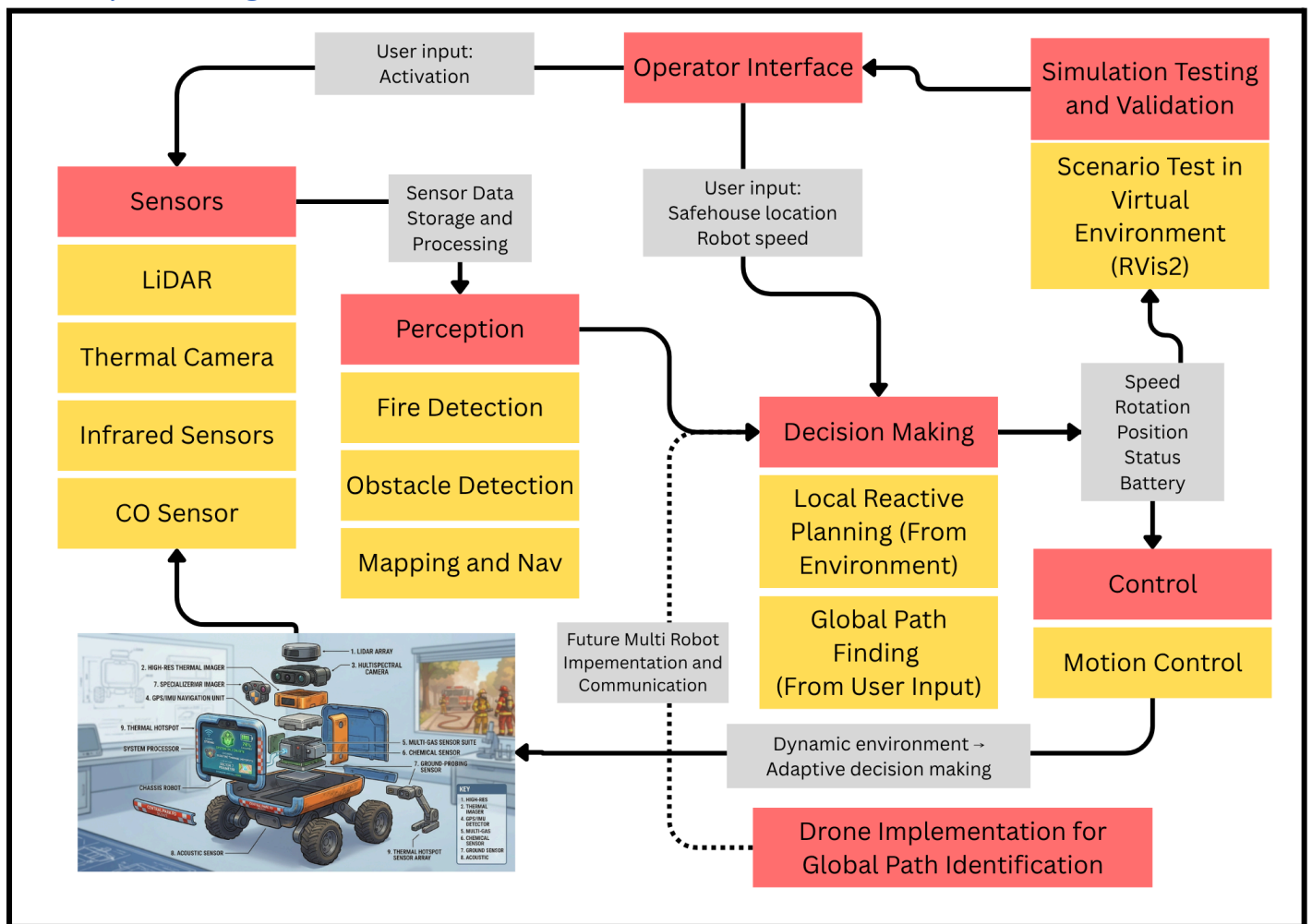
3. System Design

3.1 System Requirements

ID	Category	Requirement	Why is it important?	How will it be verified or demonstrated?	Priority: MVP or Stretch
R1	Decision making	The Husky robot shall autonomously navigate from its starting position to an operator-defined safe goal location without direct manual driving.	Provide the fundamental mobility required for environmental monitoring and rescue missions	Observe multiple waypoints and assign itself, to find the best path. The robot reaches the end goal in the simulation without direct manual driving.	MVP
R2	Perception	The system shall detect simulated fire hazards within the sensing range of the Husky and report their locations to the decision-making system.	Enables the robot to identify hazardous areas and provides meaningful perception relevant to bushfire response.	Place the fire truck with the husky robot in the middle of the forest fire, and it finds and guides the fire truck out of the forest to safety.	MVP
R3	Perception	The husky robot shall detect nearby obstacles using its simulated onboard sensors.	Prevents collisions with trees, debris, terrain and other environmental obstacles during autonomous operation.	Place obstacles along the terrain so the robot's route and demonstrate that they are detected by the perception system.	MVP
R4	Decision making	When an obstacle blocks the current route, the Husky shall avoid the obstacle and continue progressing toward the designated safe location where a traversable route exists.	Demonstrates that perception information influences autonomous decision making and improves mission safety.	Introduce an obstacle in the planned route and observe the robot stop or modify its motion to avoid collision. Furthermore, Change up the position of the fire so it has to find an alternate path.	MVP
R5	User interface	The user interface shall display the robot's current position, mission waypoints and detected fire hotspots.	Gives the operator situational awareness and allows mission progress and hazards to be interpreted quickly.	Run a mission and verify that the robot position, waypoints and detected hotspots are visibly represented on the interface.	MVP
R6	User interface	The interface shall display the current	Allows the operator to understand what the	Change the robot between mission states	Stretch

		mission status, including whether the robot is idle, navigating, responding to a hazard or has completed its mission. While also providing real time data.	autonomous system is currently doing without directly observing the robot.	and confirm that the displayed status updates appropriately.	
R7	Simulation	Simulated vegetation shall support healthy, burning and burnt states, with state transitions triggered by the fire simulation.	Creates a meaningful dynamic bushfire environment and provides perception data for the robotic system.	Trigger the simulated fire and visually demonstrate trees changing colour and/or appearance between the three states.	Stretch
R8	Perception	The system shall generate and update a map containing environmental features, robot position and identified hazardous regions.	Provides a richer representation of the disaster environment for autonomous planning and operator awareness.	Demonstrate a map being generated or updated as sensor information is received during a simulated mission.	Stretch
R9	Decision making	The system shall automatically calculate an evacuation or rescue route that avoids known fire hotspots and obstacles.	Extends the system from basic navigation to mission-level autonomous bushfire response.	Create a scenario containing hazards between the robot and destination and demonstrate generation of a safe alternative route.	MVP
R10	Multi-robot system	An aerial robot shall survey the environment and communicate detected fire or hazard locations to the ground robot.	Aerial sensing can cover a larger area quickly and improve the ground robot's situational awareness beyond its local sensors.	Demonstrate the UAV detecting a hazard and the corresponding location appearing in the ground robot's mission information or map.	Stretch

3.2 System Diagram



This system diagram shows the integration of real time environmental sensing with adaptive motion control for autonomous operations. It will be activated via an operator GUI then the onboard hardware feeds continuous data to a perception processor for fire and obstacle detection and from that navigate the environment. The decision making core will take this data and output motion control to physically execute moving functions that will navigate to a safe location. All essential system info will be stored and displayed through the GUI for testing and debugging. Secondary robots may be included and will communicate with the husky to navigate the environment faster and more efficiently. All testing will be done in the custom RViz2 environment with increasingly challenging environments.

3.3 Component Plan

Component	Purpose	Inputs	Outputs	Provided, existing, adapted, or custom?	Initial lead and support team members
Husky	Primary mobile platform used to travel through the simulated bushfire environment and investigate hazardous areas.	Velocity/motion commands; simulated environment interactions	Odometry, robot pose and sensor data	Provided	Phu

Ignition Gazebo Simulation Environment	Simulates the bushland environment, robot movement, sensors, obstacles and fire-related scenarios.	World configuration, robot commands, model states	Simulated sensor data and visual environment	Provided	Taj
Robot Sensors	Provide information about the robot's surroundings for obstacle and hazard detection. Sensors may include LiDAR and other sensors available on the provided Husky model.	Simulated environment	Range measurements, distance from goal and sensor messages	Provided and outsourced	Phu
Bushfire Environment / Tree Models	Represents vegetation and environmental obstacles within the simulated disaster area.	Initial world configuration and fire-state commands	Visual and state information for trees and obstacles	Adapted	Taj
Dynamic Fire-State System	Controls transitions of simulated vegetation between healthy, burning and burnt states.	Fire trigger, current tree state, neighbouring fire conditions or timers	Updated tree state and visual appearance	Custom	Taj
Obstacle Detection Module	Determines whether trees, debris or other objects obstruct the robot's planned path.	LiDAR/range sensor data	Obstacle locations, distance and hazard flags	Custom	Jack
Map	Maintains a representation of the robot, mission waypoints, obstacles and detected fire hazards.	Robot pose, sensor observations, hotspot locations	Updated environment/hazard map	Custom	Faiyad
Mission Manager	Coordinates the overall mission and tracks states such as idle, navigating, hazard detected and mission complete.	Operator commands, navigation status, hazard detections	Mission state, active goal and system commands	Custom	Faiyad
ROS 2 Communication Layer	Allows the different software modules to exchange sensor data, commands, mission states and detected hazards.	ROS 2 topics, services and messages	Inter-node communication	Existing Framework	Jack

3.4 Simulation and Evaluation Environment

Our team intends to create our own custom simulation environment that can properly showcase our robot's abilities and fits the scenario we have created. Some provided assets may be carried over to our custom environment particularly tree models and ground textures, however the terrain and additions required for our scenario will be implemented by us this includes hills, trenches, dense and sparse forest, fires, dead ends. As a stretch goal we may attempt to implement a dynamic fire that will spread across the sim environment which will complement and help showcase the robot's automation. We intend to use the husky as our main robot, completing the majority of the sensing and navigation itself with the stretch goal of support from an aerial drone to assist with more optimised routing by the husky to the end goal. The husky models, sensors and sim physics that are provided will be utilised in our project but will be expanded on and improved such as a more robust navigation system that takes into account obstacles and the needs of the following firetruck. Additional husky sensors will be implemented such as thermal sensors to assist the robot with our specific task and provide additional information to the operator.

3.5 Technical Risks and Early Tests

Risk or uncertainty	Why it matters	Early test or evidence needed	Mitigation or fallback plan	Owner
Path planning algorithm may fail to reliably find a safe route around dynamic fire hazards and static obstacles simultaneously	Without reliable replanning the robot cannot complete its primary mission of guiding the crew to safety	Early test of a basic global planner in a simple simulated environment with a single static obstacle, before adding dynamic fire hazards	Fall back to a simpler reactive obstacle-avoidance behaviour (e.g. stop-and-detour) if full dynamic replanning proves too complex within the timeframe	Jack
Simulated sensors (IR heat, carbon soot, LiDAR) may not provide sufficiently accurate or distinguishable data to reliably detect fire hotspots vs. normal terrain	Inaccurate detection undermines both perception and downstream decision-making	Early bench test in simulation: place known fire/hotspot regions and confirm sensor outputs correctly distinguish them from clear terrain	If custom sensor modelling is too unreliable, simplify to a known ground-truth hazard zone published directly by the simulation for perception testing	Phu
Integration between perception, decision-making and control components may introduce unexpected bugs or latency when combined	Individual components may work in isolation but fail when passing data between ROS2 nodes in real time	Early integration test connecting a minimal perception stub to the navigation stack to confirm message flow and timing before full sensor logic is added	Maintain modular ROS2 nodes with clearly defined topics/interfaces so faulty components can be isolated and replaced without breaking the whole pipeline	Jack

Custom simulation environment (dynamic fire behaviour, tree state transitions) may be difficult to build or too computationally expensive in Gazebo/Isaac	Without a working dynamic environment, several stretch goals and realistic testing are not possible	Early prototype of a small number of trees transitioning between healthy/burning/burnt states to test feasibility and performance impact	If full dynamic fire simulation is infeasible, fall back to static, pre-placed hazard zones that still test the robot's avoidance behaviour	Taj
Team's limited prior experience with ROS2 Nav2 and simulation tools may slow development	All members identified ROS2/simulation as a skill gap, which is a project delivery risk	Early individual tutorials/small tasks (e.g. each member runs a basic Nav2 demo) completed in week 1–2 to confirm baseline competency	Pair less experienced members with more experienced members for early tasks; allocate extra buffer time in the Gantt chart for ROS2-related tasks	Jack
User interface may not update in real time with accurate robot position, path and hazard data	A delayed or inaccurate UI undermines operator situational awareness, a key evaluation criterion	Early test publishing dummy position/hazard data to the UI to confirm real-time display before connecting live sensor data	If real-time updates are too complex, fall back to a periodically refreshed display (e.g. every 1–2 seconds) rather than continuous streaming	Faiyad

4. Planning and Current Evidence

4.1 Development and Integration Workflow

The team will use a shared GitHub repository, with each member working on their own feature branch (e.g., feature/perception-module). Direct commits to main are not permitted, changes must be merged via pull requests after testing and review by at least one other member, and only once the group has discussed and agreed on integration during scheduled check-ins.

Task tracking will be managed using a Kanban board (e.g., Microsoft Planner or Trello) with "To Do," "In Progress," "Testing," and "Done" columns to track progress and highlight blockers. Microsoft Teams will be used for daily communication, with key decisions documented in a shared channel for accountability.

Software interfaces between components will be defined early and documented in a shared specification, updated and communicated via Teams whenever changes are proposed.

To avoid isolated development, subsystems will be integrated into a shared development branch as soon as they are test-ready, rather than waiting until the end of semester. Weekly integration check-ins will be used to test combined functionality, catch issues early, and adjust plans accordingly.

4.2 Milestones, Gantt Chart and Immediate Next Steps

Following submission of the C1 proposal document, the team's next priority will be to establish the shared ROS 2/Gazebo development environment, Git workflow and integration process. The team will then configure the Husky platform and sensors, develop the custom simulation world, and begin scaffolding the core perception, decision-making and mission-management modules. Early integration checks will be conducted to verify communication between components and establish consistent development workflows. The team will also begin technical risk testing, focusing on sensor reliability, navigation and obstacle detection. These activities will provide the foundation for independent module testing, early identification of technical issues, and development of the first working prototype.

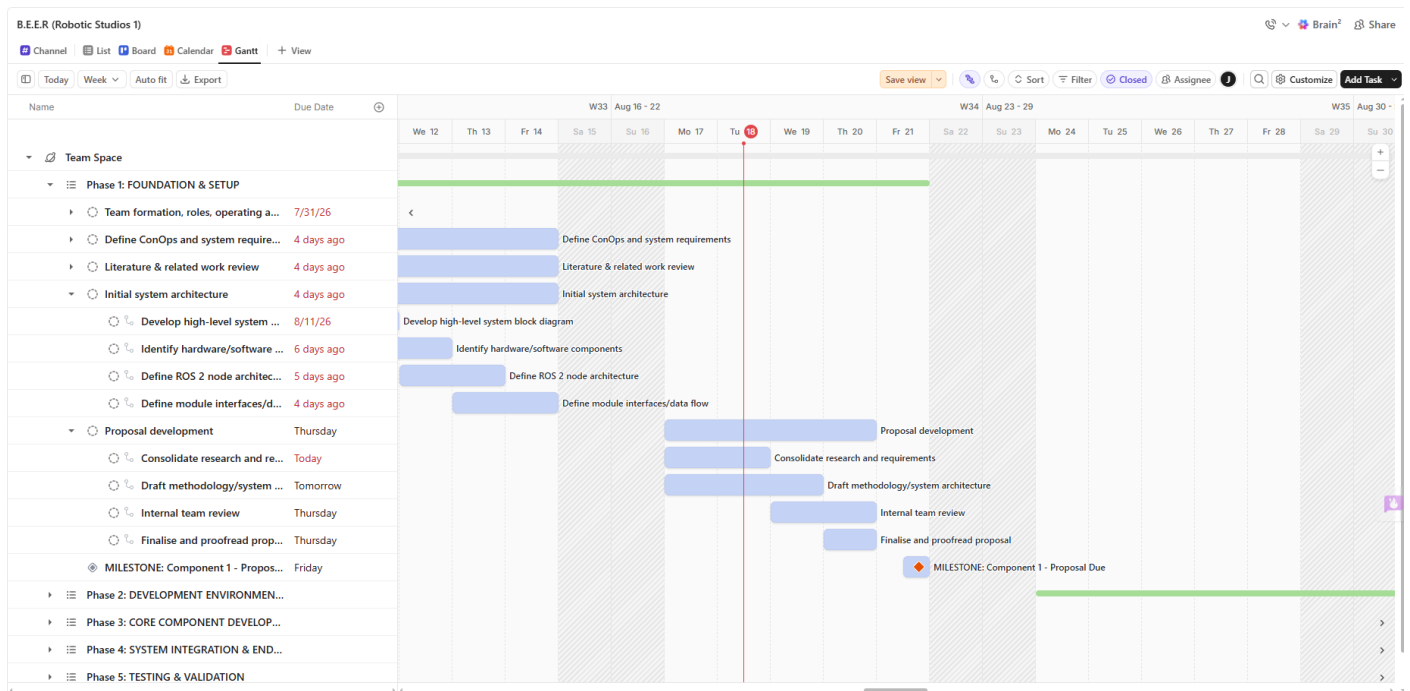


Figure 1: Snapshot of our projects Gantt Chart

4.3 Roles and Responsibilities

Activity or responsibility	Lead	Support	Expected contribution or outcome
Perception	Phu	Faiyad	Simulated sensor integration (LiDAR, camera, IR heat) for obstacle and fire detection, tuned and validated in the simulation environment
Decision-making	Faiyad	Taj	Path planning and obstacle avoidance logic enabling the Husky to navigate around hazards toward the goal location
User interface	Faiyad	Jack	Operator interface displaying robot position, path, and detected hazards in real time
Simulation development	Taj	Jack	Custom bushfire environment including tree states (healthy/burning/burnt) and fire hazard zones
System integration	Jack	Faiyad	Combining perception, decision-making, and UI components into a working end-to-end system
Testing and evaluation	Phu	Taj	Test cases for obstacle detection, path planning accuracy, and mission success criteria
Repository and software quality	All	All	Maintaining GitHub repository structure, pull request reviews, and coding standards
Stakeholder communication	All	All	Liaising with teaching staff acting as stakeholder, maintaining communication log
Project documentation	All	All	Compiling and updating the proposal and design portfolio documents
Project management	Jack	Taj	Coordinating milestones, Kanban board, and meeting schedules
Hardware/Sensor configuration	Phu	Faiyad	Configuring simulated sensor parameters (range, noise, field of view) to realistically reflect real-world Husky sensor specifications
Risk Management	Taj	Phu	Identifying, tracking, and updating technical risks and mitigation plans throughout the project
CAD/Mechanical Design	Phu	Taj	Any physical mounting or sensor placement design for the Husky

platform, modelled in SolidWorks where required

4.4 Current Technical Progress

Week 3 (11/08/2026)

The initial technical setup for the project has been completed. The provided simulated environment has been successfully loaded and configured, allowing the team to begin testing and familiarising ourselves with the Husky platform's simulated behaviour. Additionally, a shared Git repository has been established to support version control and collaborative development moving forward, ensuring that all code changes and contributions can be tracked effectively across the team.

These initial steps have laid the groundwork for the next stages of development, including implementing and testing software changes within the simulation. No major technical risks have been identified at this early stage. However, the team will continue to monitor the simulation environment closely as more complex features are integrated.

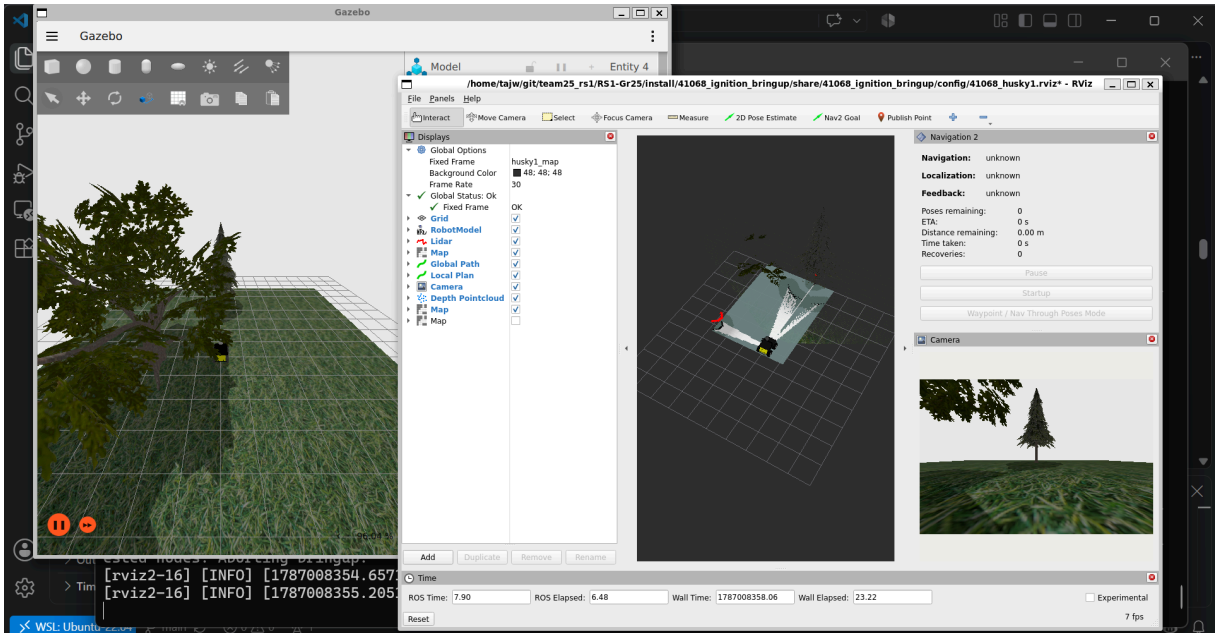


Figure 2: Simulated environment successfully loaded

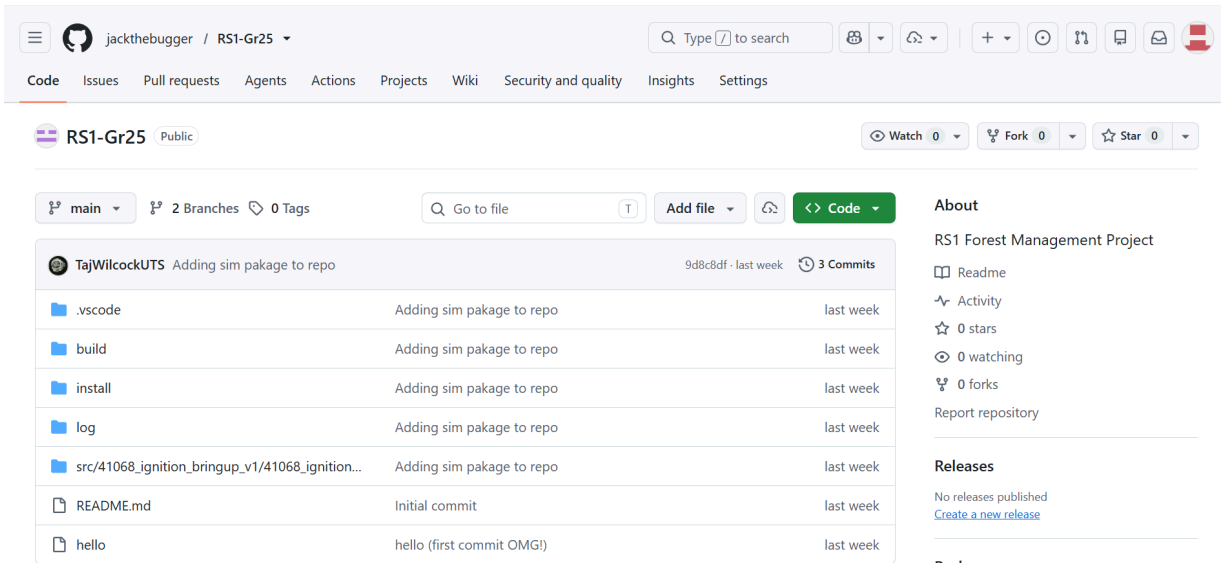


Figure 3: Screenshot of shared git repository

5. Communication

5.1 Team Decision and Action Log

Keep a concise record of important internal team discussions, decisions, changes, and actions. Focus on decisions that affect the project scope, technical approach, task allocation, or team operation. This should continue as a living record throughout the semester.

Date	Team members	Decision, issue, or discussion	Summary / outcome
11/08/2026	Taj, Faiyad, Phu, Jack	- Team roles, who's working on what - Out of the 5 ideas what was the best one - How would we approach designing our environment to show off the robots sensing/navigating capabilities - What our current goal should be	- Our current goal for the projects scope is to have one robot navigate to a set goal that has static obstacles - Then once the robot is able to show the basic sensing/self-navigating capabilities we will introduce another robot into the system as well as a dynamically changing environment
18/08/2026	Taj, Faiyad, Jack	- Discussed finalisation of C1 document - Role assignment	
	(add more rows as necessary)		

5.2 Stakeholder Communication Log

Record important discussions with the teaching staff acting as the project stakeholder. This may include both tutors and the Subject Coordinator.

Maintaining regular and constructive communication with the stakeholder is important for negotiating scope, checking assumptions, receiving feedback, and responding to changing project needs.

Date	Team members	Decision, issue, or discussion	Summary / outcome
11/08/2026	Taj, Faiyad, Phu, Jack	We presented our 5 ideas to Nadim and decided that the Bush-Fire Environmental Evacuation & Rescue (BEER)) was our best idea. Our initial idea of having the fire truck at the centre of a ring of fire and then having it navigate to a fire trail was on the right track, but was not fully showing off our robots sensor/path finding	- Included multiple obstacles into our environment that our robot has to sense (lidar, heat mapping IR, carbon soot sensor) and circumnavigate to a designated safe house - Will design the environment around our robots capabilities to better show off the robots perception and path finding capabilities - Initially we will have static fires with preset obstacles in

		capabilities. Nadim suggested we have multiple obstacles after the ring of fire as well as have one designated spot for the fire truck to aim for as opposed to the variable fire trail goal.	the world for the robot to sense and navigate (static safe paths), and then in the later stages we will introduce dynamic fires that will spread (changing paths) • For now we are going to have a robot navigate a stagnant obstacle environment, and then later on with the dynamic environment we will introduce a drone that will relay path data from above to the robot/truck (live/moving fires will change the robots path in real time)
18/08/26	Taj, Faiyad	We asked questions about a few things we were unsure about going forward into C1 and C2. We asked about if we need to change the models of the husky or the parrot, what we should have achieved by C2 and where to find programming help.	• We are not required to change the models of the husky or parrot, they can stay the same as they are in the provided package. We can change them if we want. • We should have proof that our project is coming along in C2, this includes sim screenshots which would require some implementation of sensing to find a path and modification of the sim environment. • Nadim said using the lidar in the package to navigate the environment could be difficult and as our project involves fire we could start with implementing a thermal camera. • We are allowed to use packages from the internet and genAI to help us implement and code our project - AS LONG AS WE UNDERSTAND THE CODE AND ARE ABLE TO ADJUST AND EXPLAIN IT ON THE FLY.

5.3 Priority Stakeholder Questions

Suggested length: List 2-5 substantive questions

Identify the most important questions or uncertainties that your team needs to discuss with the stakeholder before proceeding.

Questions should materially affect your scope, technical approach, feasibility, assumptions, or immediate next steps. Avoid vague questions such as "Is our idea okay?"

Discuss these questions with the teaching staff after submitting this proposal. Record the responses and explain how your team acted on them in later assessments.

Question or Uncertainty	Why is this important to clarify?	Response received Complete this in the next relevant assessment
Will our idea be sufficient to show off our robots capabilities as per criteria?	If our idea barely shows off the sensing and self-navigating aspects we will lose marks	Yes our idea is good, but we are going to need to include more obstacles for the robot to sense and navigate after the ring of fire
Do we need to include 2 robots working together right off the bat?	Allows us to work on and design something simple first instead of having to go through the troubles of implementing something advanced/complicated	No we do not need to implement 2 robots working together for the first part of the assessment, once we get the basic robot sensing and self navigation down we can then further or project with another robot
How far along do we need to be for C2?	To make sure we are not falling behind where we are expected to be up to by our stakeholder tutor.	We need to show proof that our project is coming along. This includes screenshots of a custom environment, the robot sensing and moving accordingly and avoiding obstacles while trying to reach the goal.

6. Generative AI

6.1 Generative AI Use Declaration

Declare any generative AI tools used while preparing this proposal. If none were used, state this clearly. Generative AI may support activities such as brainstorming, grammar, clarity, formatting, coding assistance, or technical explanation. However, the project direction, decisions, evidence, and submitted work must be understood and accepted by the team. Students must be able to explain their work.

Tool	Task supported	How the output was used	How the output was checked or verified
Chat GPT	Used to help brain storm for the names of our project ideas	The gen AI outputted an image for us to come up with possible names that would suit the image and by extension the project scope	We brainstormed the list of possible names as a team and ran a poll on the top 3 names we preferred, and then narrowed it down to 1, being 'Bushfire Environmental Evacuation & Rescue (B.E.E.R.)'

6.2 Planned Generative AI Protocol

Our team will use generative AI tools selectively throughout this project as a support tool, rather than as a substitute for the team's own technical development and decision-making.

Planned uses include:

- Debugging assistance and interpretation of error messages when developing ROS 2 nodes and simulation code.
- Boilerplate and scaffolding code (e.g. launch files, message/service definitions, basic node structures) that is subsequently reviewed, modified, and tested by the team.
- Drafting and refining documentation, including this proposal, meeting minutes, and later portfolio submissions.
- Explaining unfamiliar concepts, libraries, or ROS 2/Gazebo functionality to support individual learning goals.
- Generating test cases or edge-case scenarios to support evaluation of system components.

To ensure submitted work is understood and technically sound, the team will apply the following checks:

- All AI-generated or AI-assisted code will be reviewed line-by-line by at least one team member before being merged into the repository.
- Any AI-suggested technical claims, algorithms, or design recommendations will be independently verified against official documentation (e.g. ROS 2, Nav2, Gazebo) or through direct testing in simulation before being relied upon.
- Every team member must be able to explain, in their own words, any code or design decision they contribute, regardless of whether AI tools were used in its development.

AI-assisted work will be documented as it occurs, including a brief note of the tool used, the purpose of its use, and the extent of modification made to any generated output. This will be recorded in the team's decision and action log and, where relevant, in code comments or commit messages.

The team considers it inappropriate to use generative AI to fabricate test results, simulation evidence, or stakeholder communications. Additionally we also deem it inappropriate to generate entire subsystems without team members understanding their function, or to produce final report content without critical review and originality checks by the team.